GUESS THE NUMBER PRO

Linux Shell Scripting Project Report

Project	L15 – Guess The Number Pro
Student	Purvansh Darji
Environment	Ubuntu Linux / Bash
Project folder	~/linux_project/guessgame

Purpose: build a menu-driven number guessing game using Bash, file handling, validation, logging, backups, and cron scheduling.

1. Introduction and Objectives

Guess The Number Pro is a terminal-based game. Players select a difficulty, generate a random secret number, guess it using higher/lower hints, and save successful results to a high-score file.

Objectives

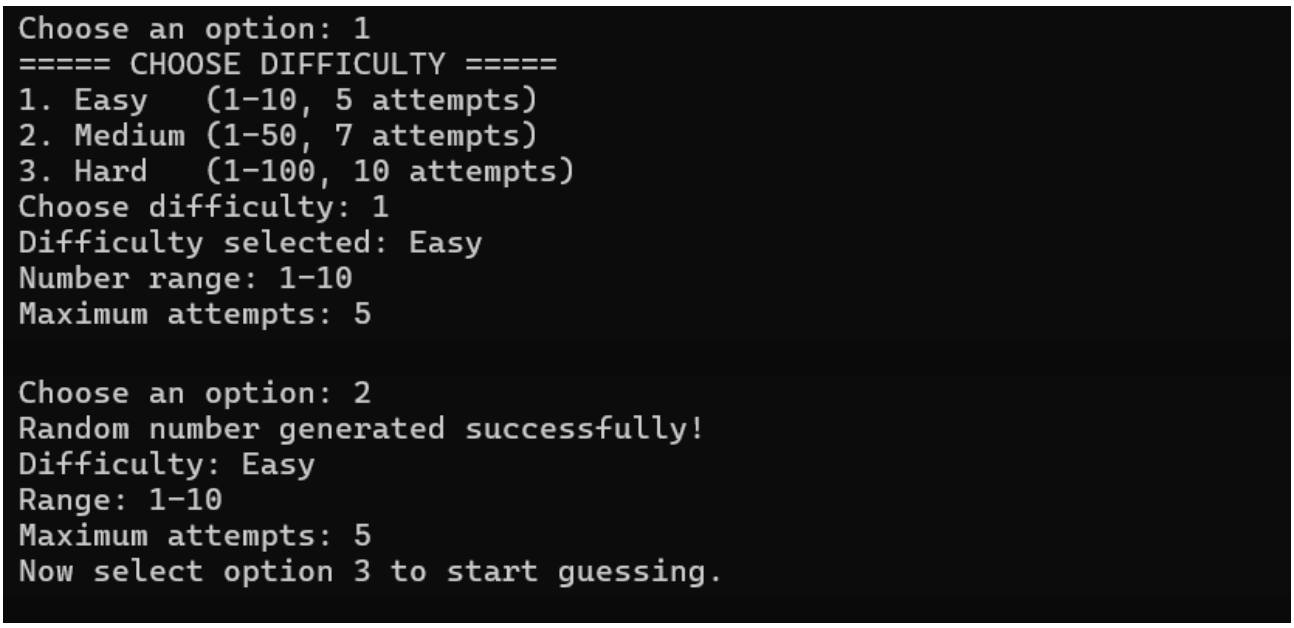
• Use Bash variables, functions, loops, conditionals, and case statements.
• Generate random numbers with $RANDOM.
• Validate user input and restrict attempts by difficulty.
• Store and display high scores in a text file.
• Use Linux commands for viewing data, creating backups, and inspecting logs.

2. Project Folder Structure

```
guessgame/
■■■ main.sh
■■■ README.md
■■■ lib/functions.sh
■■■ data/highscores.txt
■■■ reports/
■■■ backups/
■■■ logs/activity.log
```

The main.sh file provides the menu. The functions.sh file contains the feature functions. The data directory stores scores, backups contains compressed backup files, and logs records program activity.

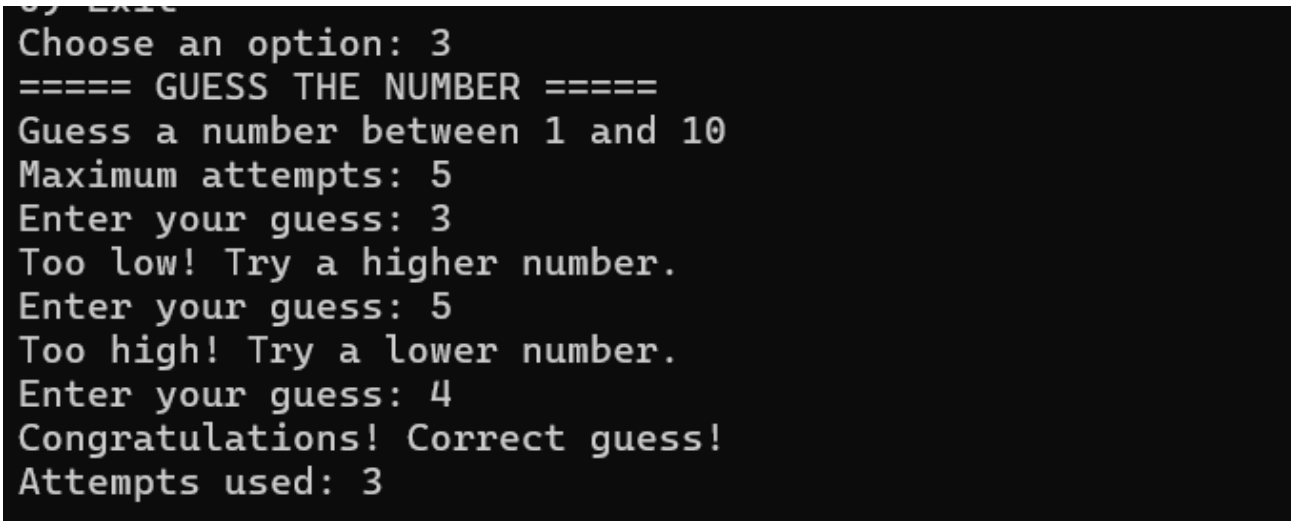
3. Feature 1 and 2: Difficulty and Random Number



Screenshot: 3. Feature 1 and 2

The player chooses Easy, Medium, or Hard. Each level defines a number range and maximum attempts. The game then generates a secret number using Bash's $RANDOM.

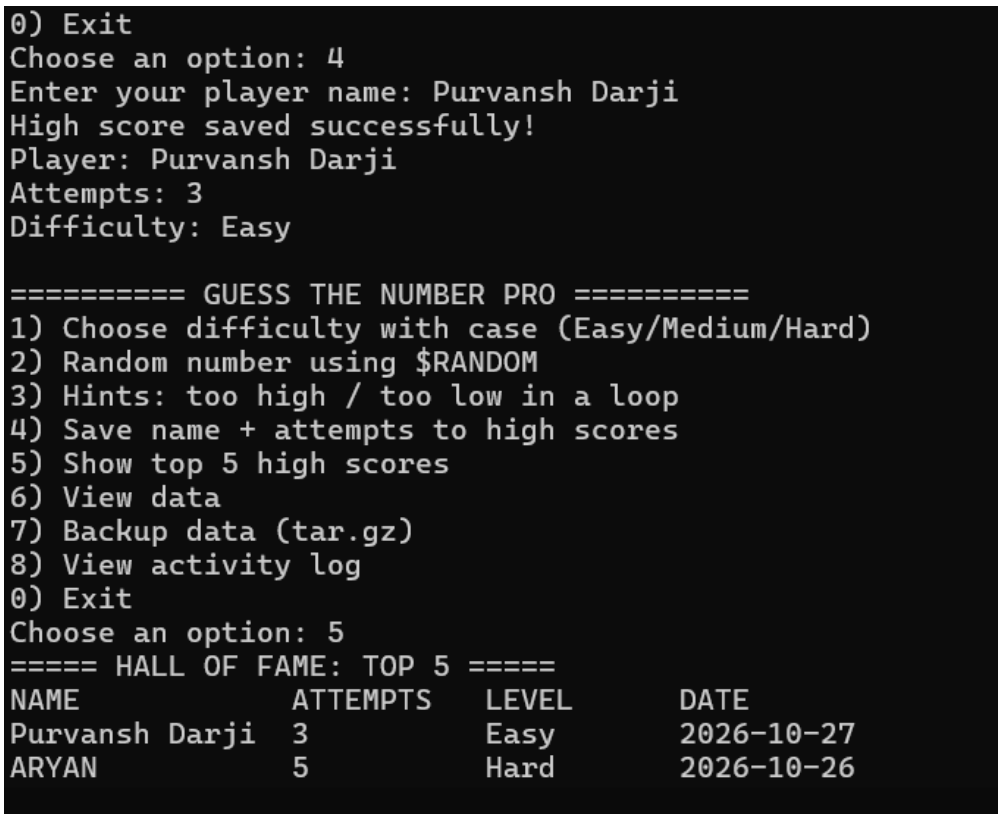
4. Feature 3: Guessing Hints



Screenshot: 4. Feature 3

The player enters guesses in a loop. The program responds with “Too high” or “Too low” until the correct number is guessed or the attempt limit is reached.

5. Feature 4 and 5: Save Score and Hall of Fame



Screenshot: 5. Feature 4 and 5

After a win, the player can save their name, attempt count, difficulty, and date. The Hall of Fame displays the saved high-score records.

6. View Data and Backup

```
===== HALL OF FAME: TOP 5 =====
NAME            ATTEMPTS   LEVEL      DATE
Purvansh Darji  3          Easy       2026-10-27
ARYAN           5          Hard       2026-10-26

========== GUESS THE NUMBER PRO ==========
1) Choose difficulty with case (Easy/Medium/Hard)
2) Random number using $RANDOM
3) Hints: too high / too low in a loop
4) Save name + attempts to high scores
5) Show top 5 high scores
6) View data
7) Backup data (tar.gz)
8) View activity log
0) Exit
Choose an option: 6
     1  name,attempts,level,date
     2  ARYAN,5,Hard,2026-10-26
     3  Purvansh Darji,3,Easy,2026-10-27

========== GUESS THE NUMBER PRO ==========
1) Choose difficulty with case (Easy/Medium/Hard)
2) Random number using $RANDOM
3) Hints: too high / too low in a loop
4) Save name + attempts to high scores
5) Show top 5 high scores
6) View data
7) Backup data (tar.gz)
8) View activity log
0) Exit
Choose an option: 7
Backup saved: backups/data_20261027_150240.tar.gz (4.0K)
```

Option 6 displays the high-score CSV/text data. Option 7 creates a compressed .tar.gz backup. The tar command is useful for packaging project data for storage or submission.

7. Activity Log and Validation

```
Choose an option: 8
2026-10-27 14:55:29 | Random number generated for Medium
2026-10-27 14:56:21 | Game lost after 7 attempts
2026-10-27 14:56:40 | Score not saved: game not won
2026-10-27 14:56:43 | Top 5 high scores displayed
2026-10-27 14:57:11 | Top 5 high scores displayed
2026-10-27 14:57:14 | backup created backups/data_20261027_145714.tar.gz
2026-10-27 14:59:30 | Difficulty selected: Easy
2026-10-27 14:59:35 | Game failed: secret number not generated
2026-10-27 14:59:57 | Difficulty selected: Easy
2026-10-27 15:00:03 | Random number generated for Easy
2026-10-27 15:00:23 | Random number generated for Easy
2026-10-27 15:01:07 | Game won in 3 attempts
2026-10-27 15:01:31 | High score saved for Purvansh Darji
2026-10-27 15:01:40 | Top 5 high scores displayed
2026-10-27 15:02:40 | backup created backups/data_20261027_150240.tar.gz
```

The activity log records events such as difficulty selection, random number generation, game results, score saving, leaderboard display, and backup creation. Input validation helps prevent invalid values and avoids saving a score for a lost game.

8. Cron Task and Submission Checklist

The project guide requires a daily 9:00 AM cron task that writes the current top five scores to hall_of_fame.txt. The cron setup and successful run evidence have not yet been supplied for this report, so they must be verified and added before final submission.

Commands to verify the cron task

```
./main.sh --cron
crontab -l
cat logs/cron.log
```

Final submission checklist

• Verify all menu options and the cron task.
• Add a screenshot of crontab -l and logs/cron.log after a successful run.
• Create and verify C44_L15_guessgame.tar.gz.
• Submit this report together with the project archive.

9. What I Learned

I practised writing Bash functions, using case statements and loops, generating random numbers, validating inputs, working with text files, displaying sorted records, creating tar.gz backups, and reading activity logs. I also learned that scheduled tasks must be tested and their output checked.

Conclusion

Guess The Number Pro demonstrates a basic menu-driven Linux application with game logic and file operations. The project is ready for final verification once the scheduled cron task is tested and documented.